

202721

archisha - incode

~~1 2 3 4 5 6 7 1 0 5 7 3 8 9~~
~~149 1024 94096~~

1 2 4 8 16 32 64 128 256
 a r c h i s h a i n c o d e

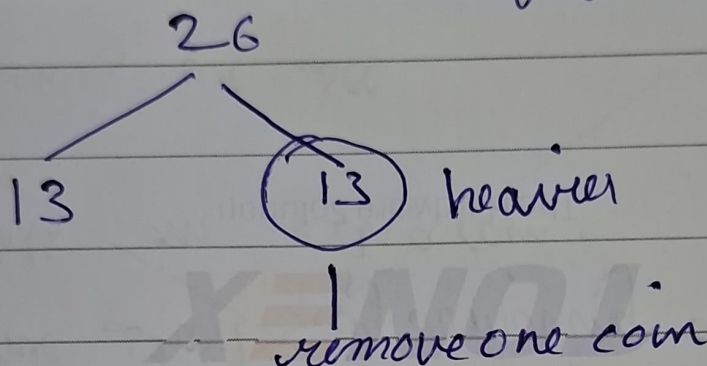
1 18 38 9198 1 0 9 14 3545

↓ ↓ ↓ ↓ ↓ ↓ ↓
 0 512 2048 8192

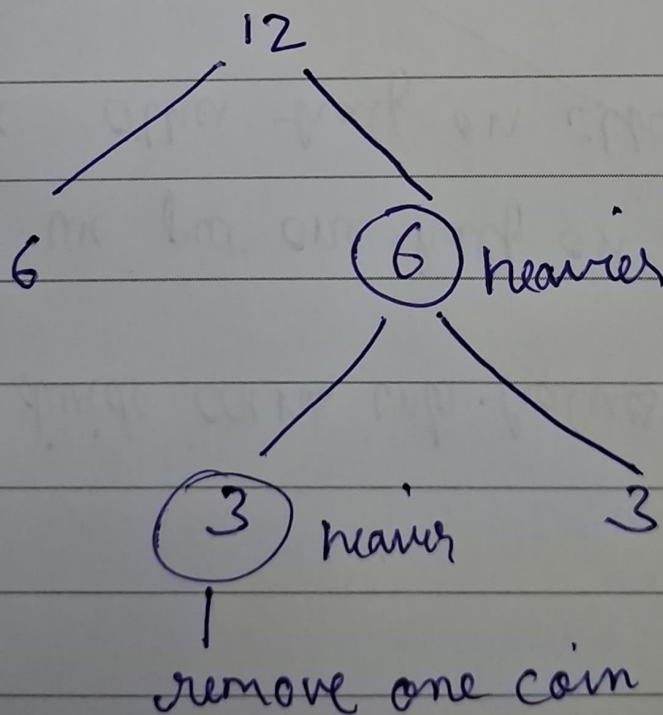
1 (36)
 ↓
 (12)
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 72
 ↓
 (64)

16384

- 1) Guess : 13
- 2) For 27 coins, ~~26~~ take out one coin, if then divide the 26 coins into half, if the scale weighs equal, the coin u just removed is heavier. If not, we proceed like this (without loss of generality)



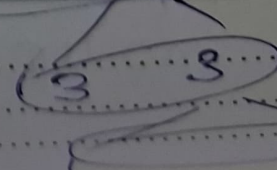
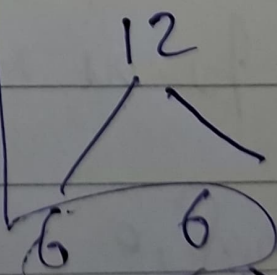
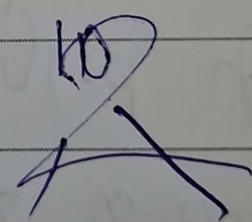
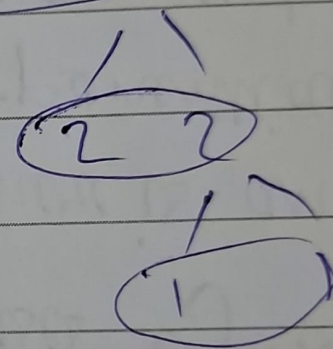
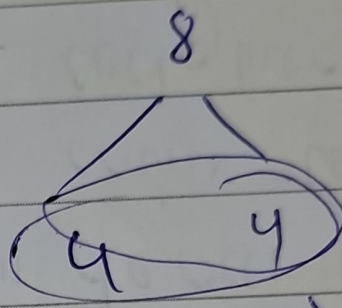
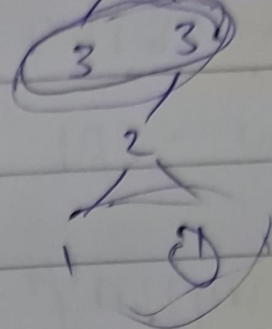
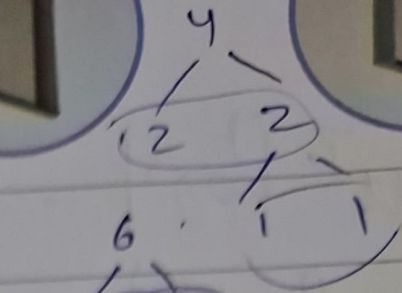
∴ In worst case, it takes 4 weighings to find the ~~defected~~ fake coin.



Date:

3) Table for fewest weighings

2	1	18	4
3	1	19	4
4	2	20	4
5	2	21	4
6	3 2	22	4
7	3 2	23	4
8	4 3	24	4
9	4 3	25	4
10	3	26	4
11	4 3	27	4
12	4 3		
13	4 3		
14	3		
15	3		
16	4		
17	4		



Date:

4) Observation : The no. of weighings go up by 1 when your 'n' becomes of the form 2^q .

5) Fewest weighings for $n = \lfloor \log n \rfloor$

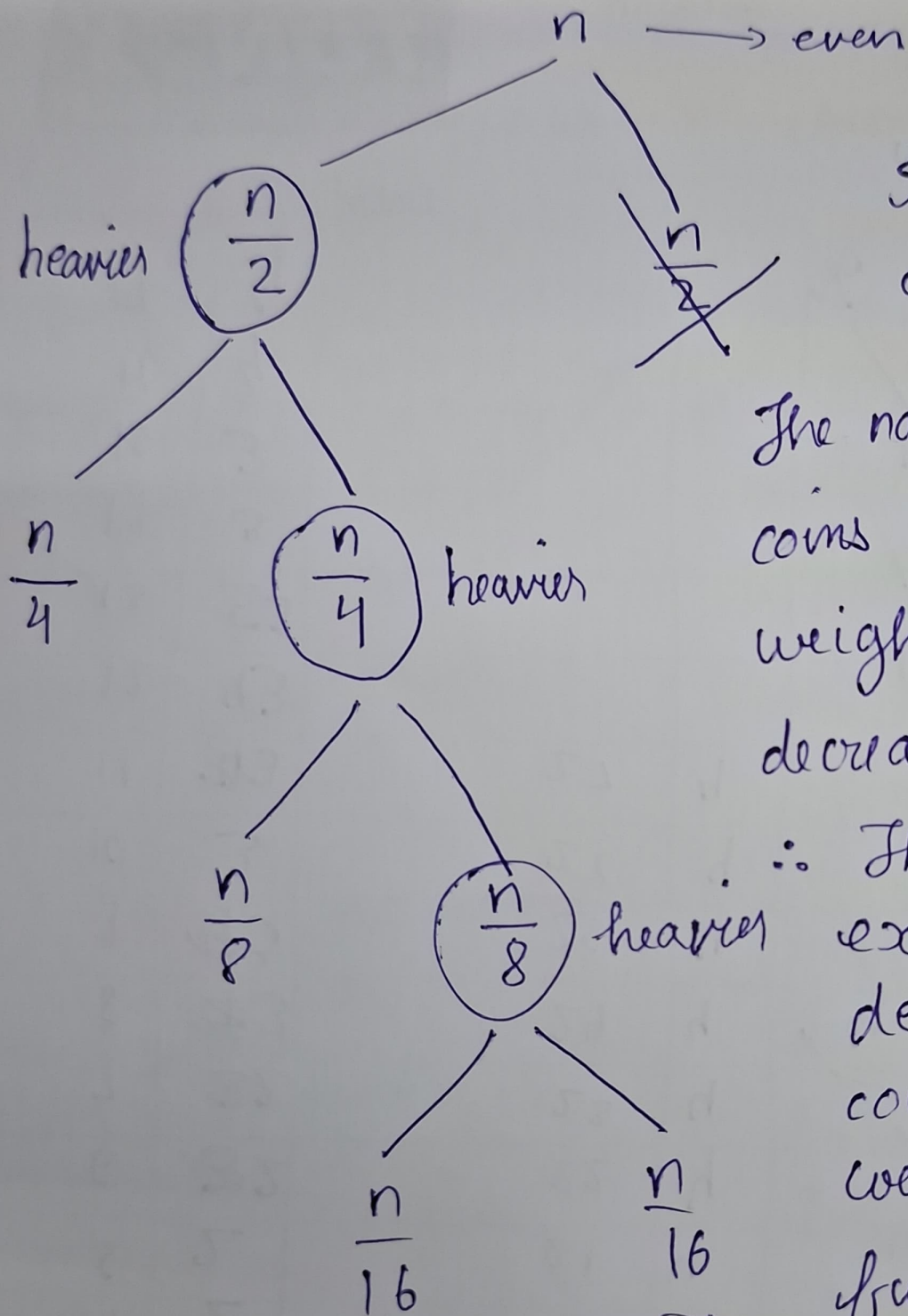
* If no. of coins is even, divide them into half.

If no. of coins ~~is~~ to be weighed is odd,

remove one coin.

coin which just removed
was fake
(^{desired} coin found)

If no. of coins left is even
~~divide~~ eliminate
the lighter half,
divide the heavier
half into half
again,
weigh it, and
repeat the process
accordingly.



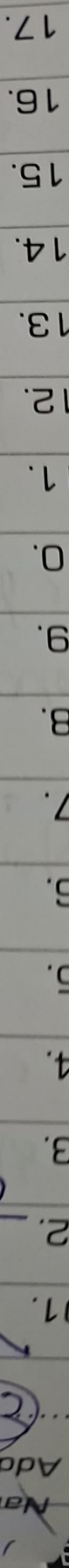
In our algorithm,

The no. of coins to be weighed further decreases by half.

$\therefore$ There is an exponential decrement of coins to be weighed further

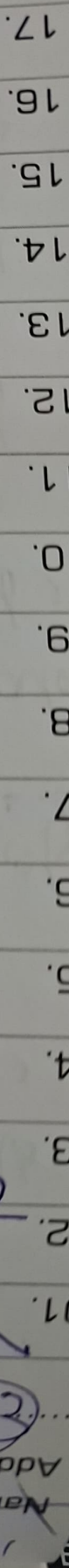
$\therefore$ No. of weighings
 $= \lfloor \log_2 n \rfloor$

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6.
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1.

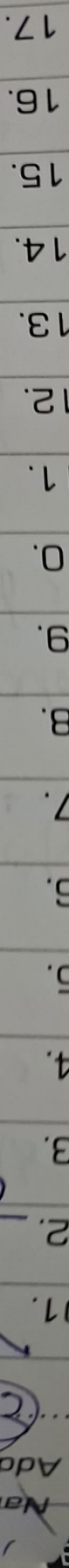


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